CSE387 - Data Mining

Phase 1

Team 7

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Table of Contents

1.0 Introduction 2

2.0 Dataset Understanding 2

3.0 Preprocessing.....5

4.0 Outlier Detection 6

5.0 Data Assessment and Correlation Analysis 8

6.0 Encoding 12

7.0 Conclusion.....13

1.0 Introduction

The objective of this phase is to analyze the given dataset and prepare it for machine learning models to perform accurate predictions based on the given features. This involves understanding the structure of the data, identifying potential issues, and applying appropriate preprocessing techniques. The dataset consists of a categorical mushroom dataset containing 8,124 observations and 23 feature dimensions, with the goal of building a foundation for robust classification models.

Key Phase 1 Activities:

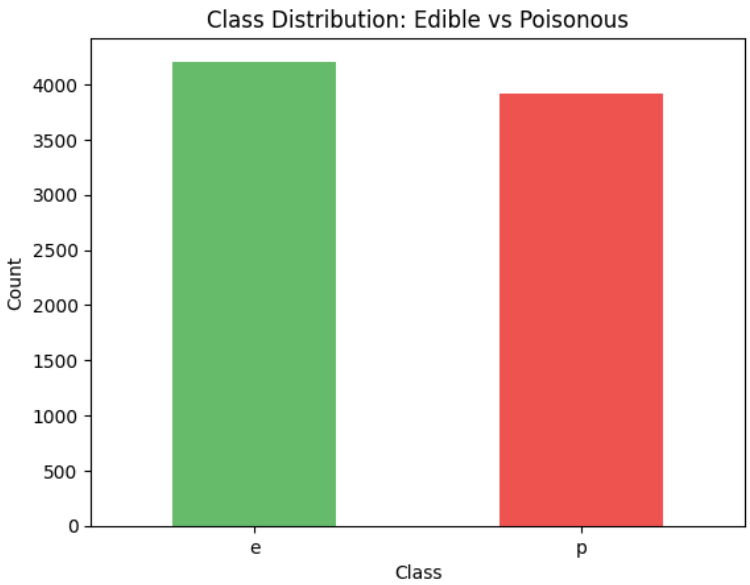
- **Data Exploration:** Load and inspect the raw dataset to understand structure, feature types, and target distribution.
- **Preprocessing:** Handle missing values, remove duplicates, and validate data integrity.
- **Outlier Detection:** Identify and flag rare categorical patterns using frequency based statistical thresholds.
- **Association Analysis:** Quantify relationships between features and the target using categorical association measures.
- **Encoding Preparation:** Create a leakage-safe, encoding-ready representation of the dataset.

2.0 Dataset Understanding

The dataset includes 8,124 mushroom samples, each described by 23 categorical attributes representing physical characteristics of mushrooms. All variables in the dataset are categorical. No continuous numerical variables are present, which require categorical preprocessing techniques before machine learning models can be applied.

The target variable **class** indicates whether a mushroom is e (edible) or p (poisonous):

Class	Count
Edible (e)	4208
Poisonous (p)	3916



2.1 Feature Space

The dataset contains **23 categorical predictor variables** describing mushroom morphology and habitat conditions. These features represent observable characteristics such as cap shape, gill structure, stalk properties, and habitat environment.

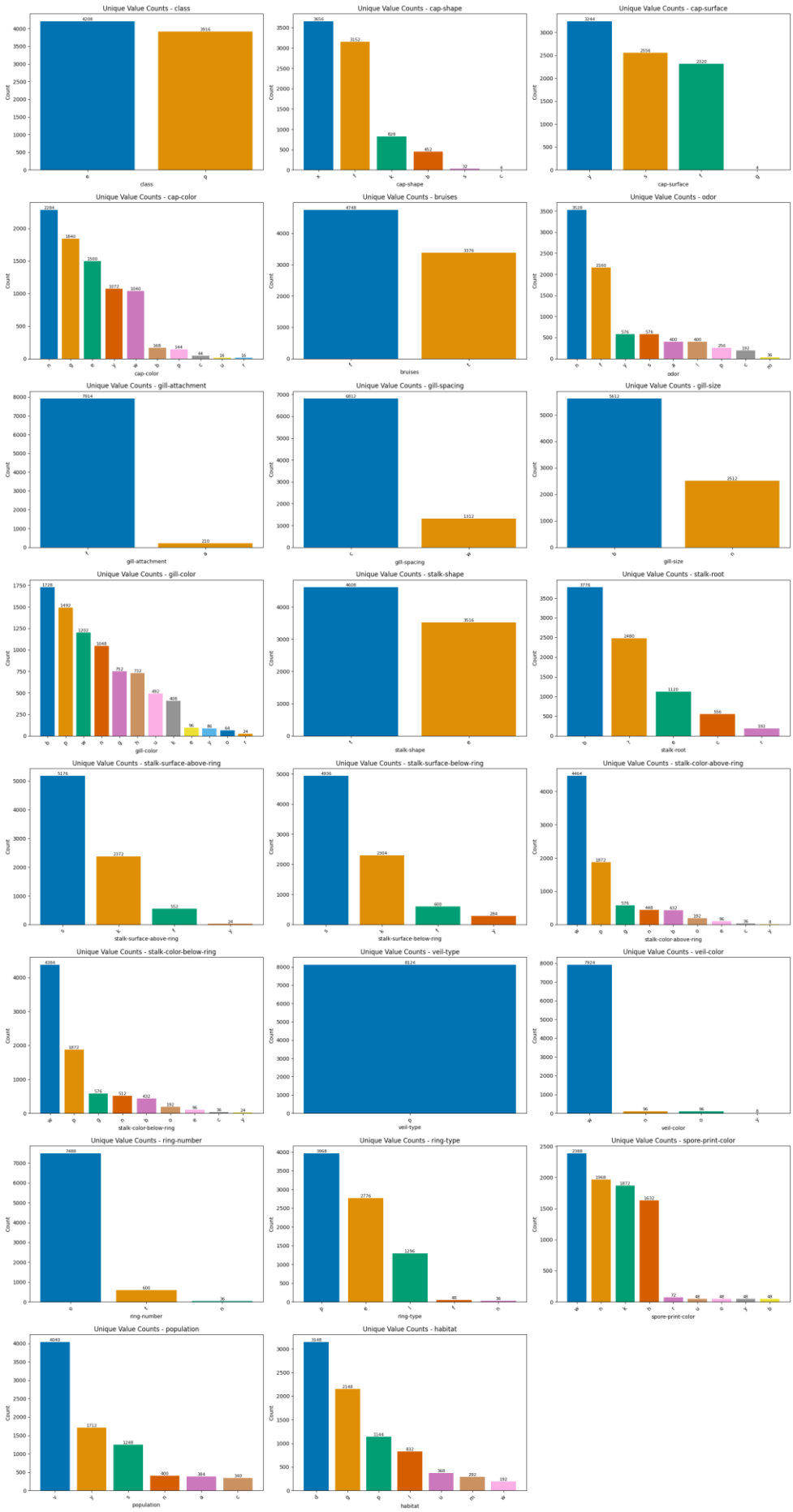
```
Shape: (8124, 23)

General information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8124 entries, 0 to 8123
Data columns (total 23 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   class                     8124 non-null   object
 1   cap-shape                 8124 non-null   object
 2   cap-surface               8124 non-null   object
 3   cap-color                 8124 non-null   object
 4   bruises                   8124 non-null   object
 5   odor                      8124 non-null   object
 6   gill-attachment           8124 non-null   object
 7   gill-spacing              8124 non-null   object
 8   gill-size                 8124 non-null   object
 9   gill-color                8124 non-null   object
 10  stalk-shape               8124 non-null   object
 11  stalk-root                8124 non-null   object
 12  stalk-surface-above-ring  8124 non-null   object
 13  stalk-surface-below-ring  8124 non-null   object
 14  stalk-color-above-ring    8124 non-null   object
 15  stalk-color-below-ring    8124 non-null   object
 16  veil-type                 8124 non-null   object
...
 21  population                8124 non-null   object
 22  habitat                   8124 non-null   object
dtypes: object(23)
memory usage: 1.4+ MB
```

Examples of feature groups include:

- **Cap features:** shape, surface, color
- **Gill features:** attachment, spacing, size, color
- **Stalk features:** root type, surface texture, color
- **Other characteristics:** veil type, ring type, spore print color, habitat

Each category is encoded using short symbolic values (e.g., *n = brown*, *y = yellow*, *w = white*).


Unique Value Counts - class
Unique Value Counts - cap-shape
Unique Value Counts - cap-surface
Unique Value Counts - cap-color
Unique Value Counts - bruises
Unique Value Counts - odor
Unique Value Counts - gill-attachment
Unique Value Counts - gill-spacing
Unique Value Counts - gill-size
Unique Value Counts - gill-color
Unique Value Counts - stalk-shape
Unique Value Counts - stalk-root
Unique Value Counts - stalk-surface-above-ring
Unique Value Counts - stalk-surface-below-ring
Unique Value Counts - stalk-color-above-ring
Unique Value Counts - stalk-color-below-ring
Unique Value Counts - veil-type
Unique Value Counts - veil-color
Unique Value Counts - ring-number
Unique Value Counts - ring-type
Unique Value Counts - spore-print-color
Unique Value Counts - population
Unique Value Counts - habitat
Count

3.0 Preprocessing

Data preprocessing ensures the dataset is clean and suitable for analysis. Since the mushroom dataset consists entirely of categorical variables, preprocessing focuses on data quality rather than numerical transformations. The main preprocessing steps include:

- Handling missing values
- Removing redundant features
- Checking for duplicate records

These steps ensure consistency and reliability before further statistical analysis.

3.1 Missing Value

Missing values in the dataset are encoded using the symbol “?”. Initial inspection shows that missing values are concentrated in the **stalk-root** feature, while all other features contain complete observations.

Since all variables are categorical, missing values are replaced using **mode imputation** (the most frequent category). This method preserves the natural distribution of categories within each feature. For the **stalk-root** feature, the most frequent category was **b**, which was used to replace the missing values.

```
Missing values per column before filling:

stalk-root                  2480
stalk-surface-above-ring       0
population                     0
spore-print-color              0
ring-type                      0
ring-number                    0
veil-color                     0
veil-type                      0
stalk-color-below-ring         0
stalk-color-above-ring         0
stalk-surface-below-ring       0
class                          0
cap-shape                      0
stalk-shape                    0
gill-color                     0
gill-size                      0
gill-spacing                   0
gill-attachment                0
odor                           0
bruises                        0
cap-color                      0
cap-surface                    0
habitat                        0
dtype: int64
```

stalk-root counts BEFORE fill:

stalk-root	count
b	3776
NaN	2480
e	1120
c	556
r	192

stalk-root counts AFTER fill:

stalk-root	count
b	6256
e	1120
c	556
r	192

3.3 Feature Simplification

During inspection, the feature **veil-type** was found to contain only a single category across all observations. Features with a single constant value do not contribute useful information for analysis or machine learning models. Therefore, this feature was removed from the dataset.

```
Dropped features with single category: ['veil-type']
```

3.4 Duplicate Records

Duplicate observations were checked and removed to avoid redundant information. The inspection revealed no duplicate rows, indicating that all records represent unique observations.

```
Dropped 0 duplicate rows. Remaining rows: 8124
```

4.0 Outlier Detection

Traditional outlier detection methods such as **z-score** or **IQR** are designed for numerical data and rely on distance or magnitude comparisons. Since the mushroom dataset contains only **categorical variables**, outliers are instead defined as observations containing **rare category values**.

Rare categories represent values that occur very infrequently in the dataset and may indicate unusual biological variants, data recording errors, or observations from uncommon environmental conditions.

4.1 Frequency Based Detection

Rare Category Threshold

Outliers are identified using **frequency thresholding**. A category is considered rare if its frequency is less than **1% of the dataset size**. This threshold captures categories that appear extremely infrequently while keeping common patterns intact.

Detection Procedure

For each feature (excluding the target variable):

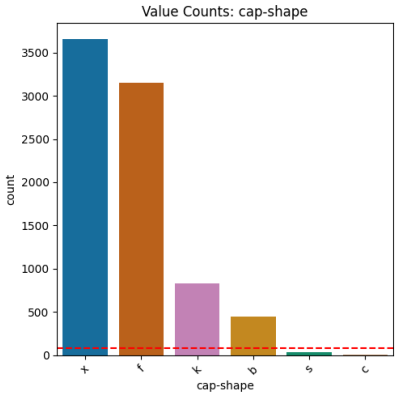
1. Compute the frequency of all category values
2. Identify categories occurring below the threshold
3. Mark rows containing at least one rare category as outliers

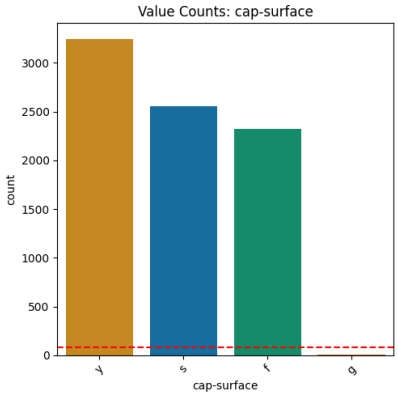
The final outlier set consists of the union of all rows containing rare categories across any feature.

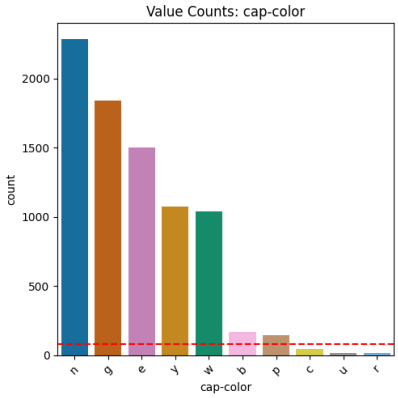
Rare Category Detection Results

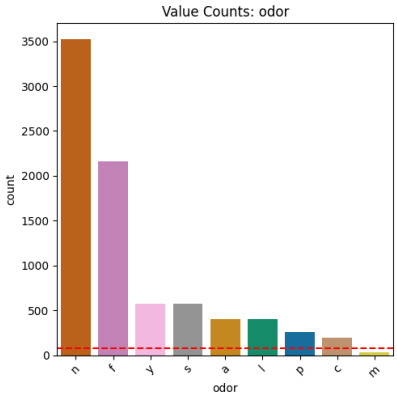
```
Global Outliers Found: 473 rows
```

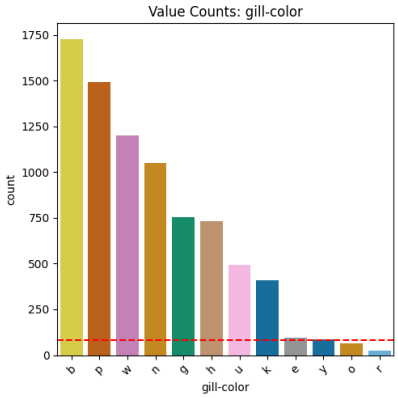
	Feature	Rare Values	Total Rows Affected
0	cap-shape	[s, c]	36
1	cap-surface	[g]	4
2	cap-color	[c, u, r]	76
3	odor	[m]	36
4	gill-color	[o, r]	88
5	stalk-surface-above-ring	[y]	24
6	stalk-color-above-ring	[c, y]	44
7	stalk-color-below-ring	[c, y]	60
8	veil-color	[y]	8
9	ring-number	[n]	36
10	ring-type	[f, n]	84
11	spore-print-color	[r, u, o, y, b]	264

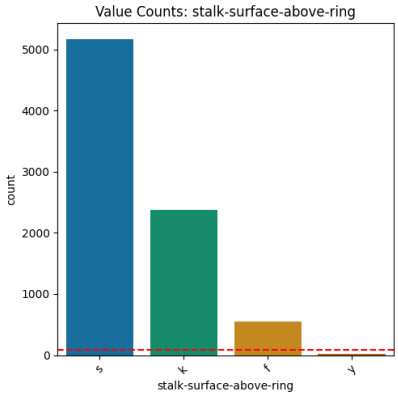

Value Counts: cap-shape
count
cap-shape
x f k b s c

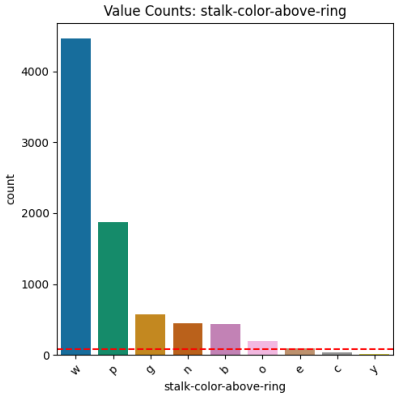

Value Counts: cap-surface
count
cap-surface
y s f g

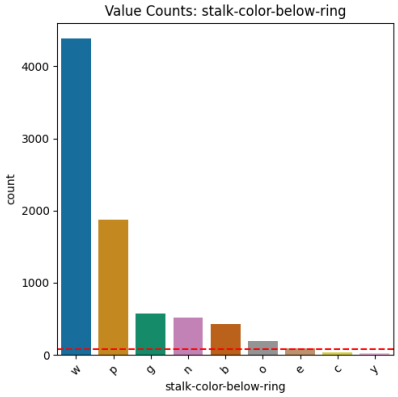

Value Counts: cap-color
count
cap-color
n g e y w b p c u r

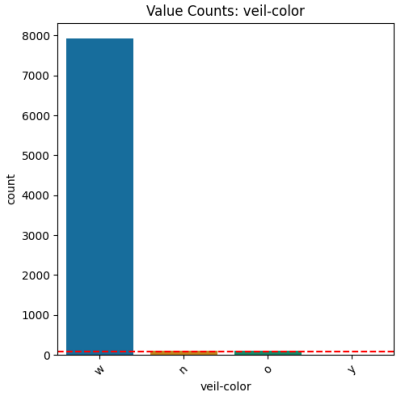

Value Counts: odor
count
odor
n f y s a l p c m

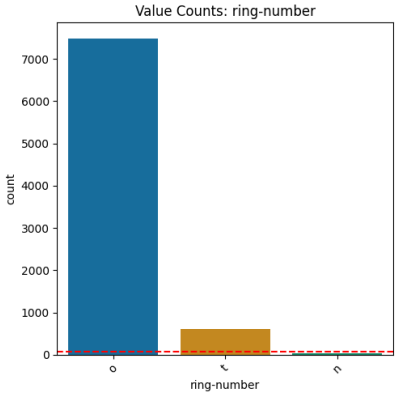

Value Counts: gill-color
count
gill-color
b p w n g h u k e y o r

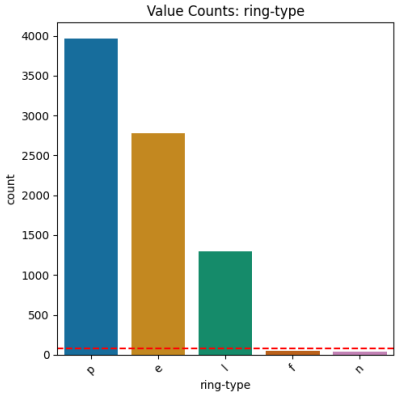

Value Counts: stalk-surface-above-ring
count
stalk-surface-above-ring
s k f y

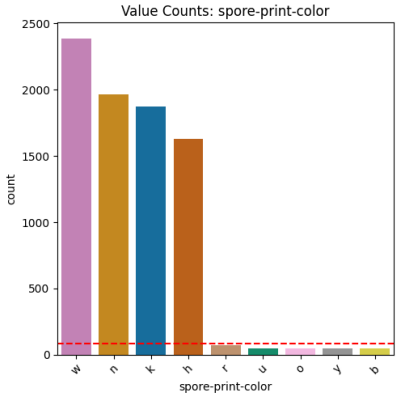

Value Counts: stalk-color-above-ring
count
stalk-color-above-ring
w p g n b o e c y


Value Counts: stalk-color-below-ring
count
stalk-color-below-ring
w p g n b o e c y


Value Counts: veil-color
count
veil-color
w n o y

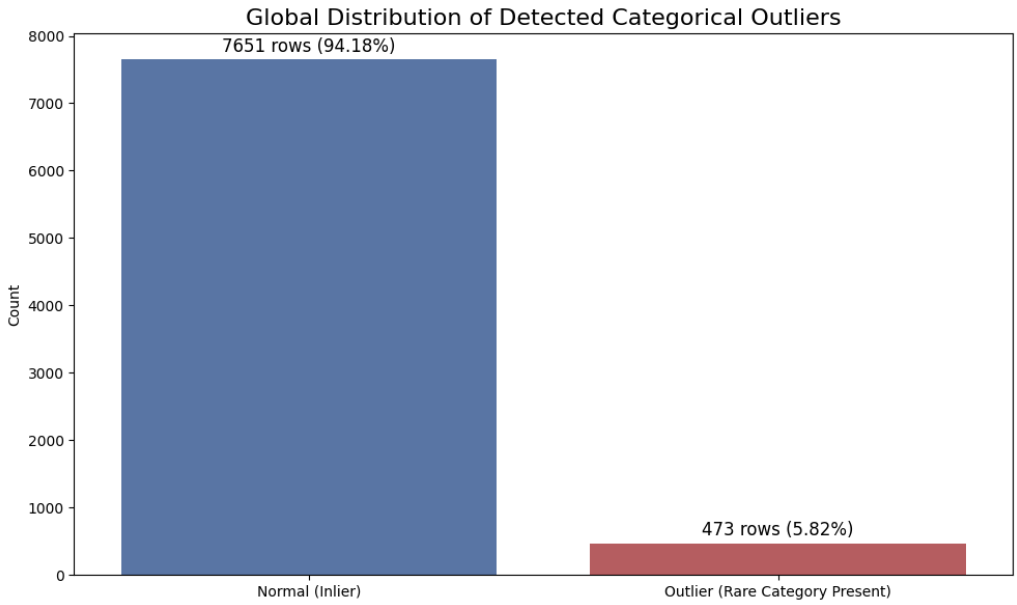

Value Counts: ring-number
count
ring-number
o t n


Value Counts: ring-type
count
ring-type
p e l f n


Value Counts: spore-print-color
count
spore-print-color
w n k h r u o y b

4.2 Outlier Summary

The overall results of the outlier detection process are summarized below.



Rows containing rare categories were flagged as outliers and excluded from subsequent statistical analysis. Removing these observations reduces noise and prevents extremely sparse categories from distorting feature association measurements.

5.0 Data Assessment and Correlation Analysis

After completing data preprocessing and removing outliers, the next step is to analyze the relationships between categorical features. Understanding these relationships is critical for identifying dependencies, reducing redundancy, and determining which features contribute most to predicting the target variable (class).

The correlation analysis was performed on the cleaned dataset after removing all rows flagged as outliers.

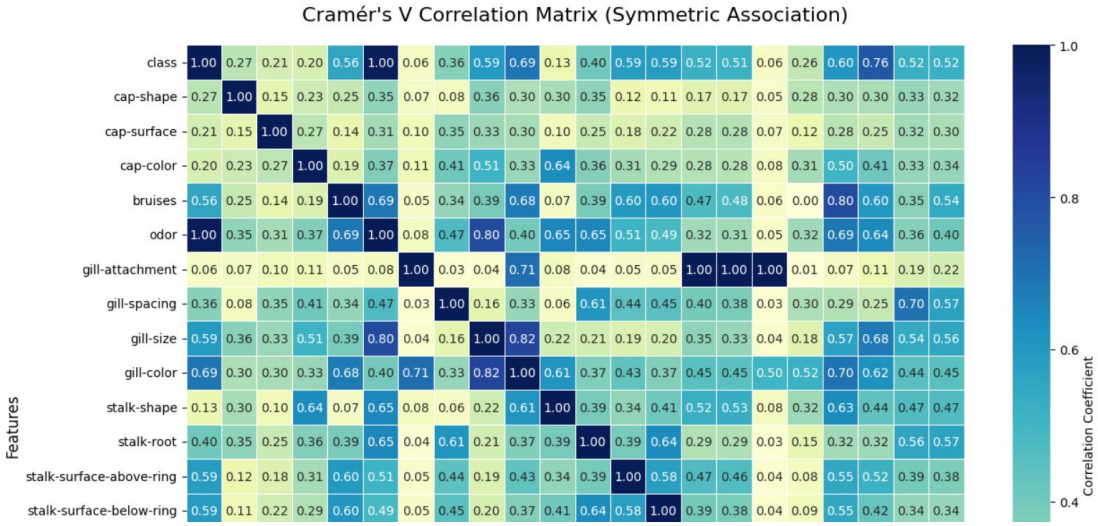
- **Total Rows:** 7,651
- **Total Features:** 22 (excluding the outlier flag)

This ensures that the computed associations are not distorted by rare or noisy categories identified in the previous step.

5.1 Cramér's V Symmetric Association Analysis

Cramér's V is based on the Chi-Square statistic and measures the strength of association between two categorical variables. It produces values between 0 and 1 where 0 implies **No Association** and 1 implies **Perfect Association**

A full pairwise matrix was computed for all feature combinations using contingency tables.



Key Findings

The strongest feature associations identified were:

- gill-attachment ↔ veil-color → 1.000
- gill-attachment ↔ stalk-color-above-ring → 1.000
- gill-attachment ↔ stalk-color-below-ring → 1.000
- class ↔ odor → 0.995
- gill-size ↔ gill-color → 0.822
- odor ↔ gill-size → 0.805
- bruises ↔ ring-type → 0.803

Several important insights can be derived:

1. Perfect Correlations (V = 1.0)

Features such as *gill-attachment*, *veil-color*, and *stalk-color* are perfectly correlated. This suggests redundancy, meaning one feature may be removed without losing information.

2. Strong Relationship with Target Variable

Odor has an extremely strong association with *class* (0.995), indicating it is a critical predictor.

3. Clustered Feature Groups

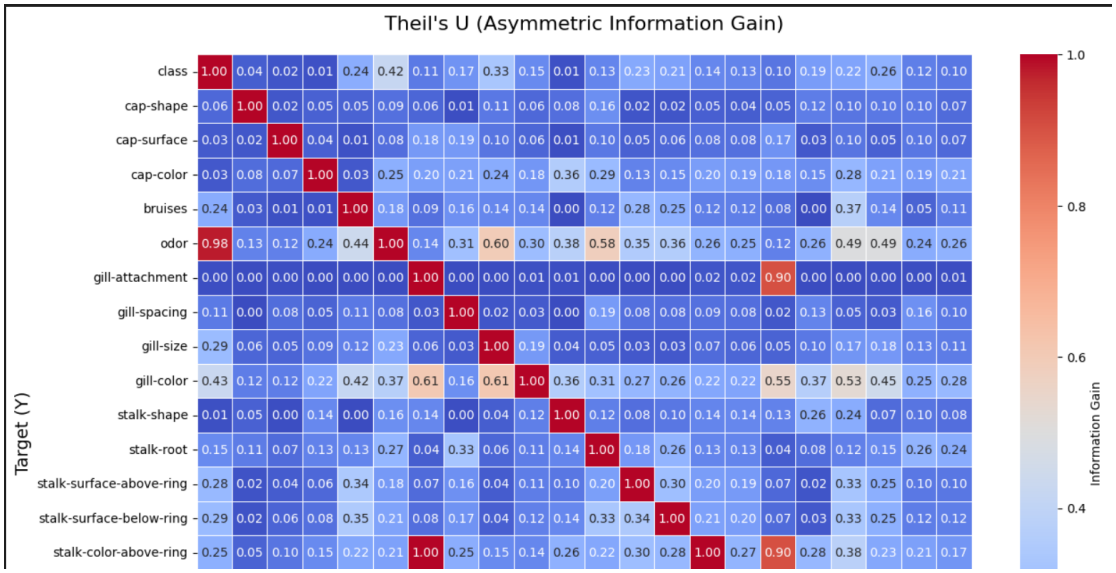
Features like *gill-size*, *gill-color*, and *odor* form tightly connected groups, suggesting underlying structural relationships.

5.2 Theil's U Asymmetric Information Gain

Theil's U measures how much knowing one variable reduces uncertainty about another. This is particularly useful for identifying predictive features for the target variable.

$$U(X \rightarrow Y) \neq U(Y \rightarrow X)$$

Values range from 0 to 1 where 0 implies **no predictive power** and 1 implies **perfect prediction**



Predictive Insights (Information Gain > 50%)

The following features provide strong predictive power:

- odor → class → 97.81%
- ring-type → bruises → 56.23%
- stalk-color-above-ring → gill-attachment → 100%
- population → gill-spacing → 50.43%
- gill-color → gill-size → 61.08%
- spore-print-color → ring-type → 54.27%

Interpretation

1. Dominant Predictor: Odor

Odor alone explains ~98% of the uncertainty in mushroom toxicity making it the most influential feature in the dataset.

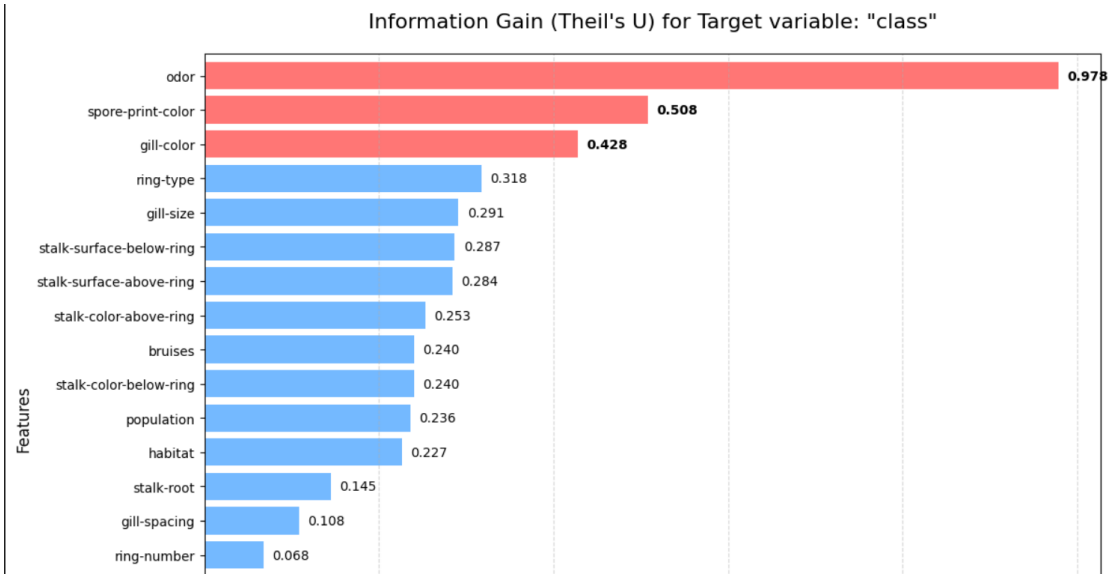
2. Deterministic Relationships

Some features (e.g., stalk-color-above-ring → gill-attachment) show 100% information gain, indicating direct dependency.

3. Moderate Predictors

Features such as *spore-print-color* and *population* provide moderate predictive value and may enhance model performance when combined.

5.3 Target-Specific Feature Importance



To better understand prediction of the target variable (class), features were ranked using Theil's U.

Top Informative Features for Mushroom Toxicity:

1. Odor → 97.81%
2. Spore-print-color → 50.78%
3. Gill-color → 42.76%
4. Ring-type → 31.78%
5. Gill-size → 29.13%

From the correlation and information gain analysis:

- The dataset contains highly correlated feature groups, indicating redundancy.
- Odor is the single most important predictor of mushroom toxicity.
- Several features exhibit deterministic or near-deterministic relationships, which may impact model design.
- Removing redundant features can reduce dimensionality, improve model efficiency and prevent overfitting

6.0 Encoding

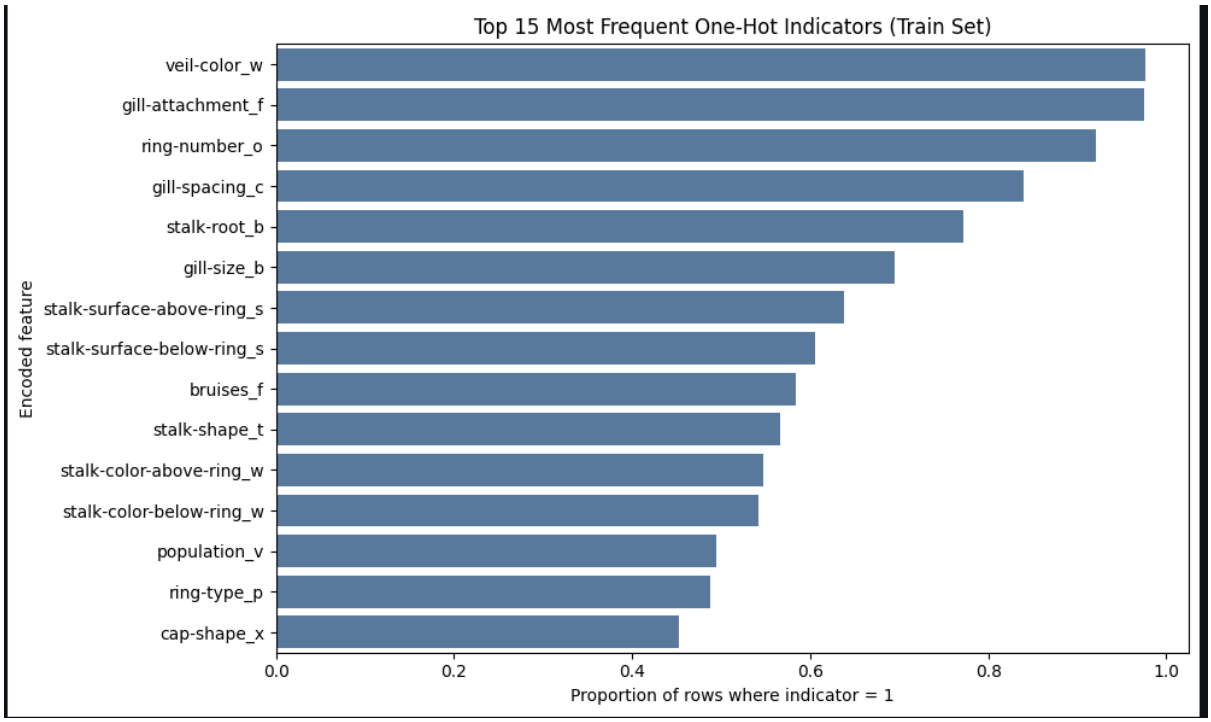
Machine learning algorithms operate on numeric feature spaces and require numeric target variables. While statistical analysis worked directly with categorical data, Phase 2 model training requires converting all categorical variables to numeric representations. The encoding strategy must preserve information while enabling efficient computation.

The encoding pipeline consists of two components:

- **Target variable encoding:** The binary target class is encoded as e (edible) = 0, p (poisonous) = 1
- **Feature encoding:** All categorical predictor variables are transformed using **one-hot encoding**, where a feature with kcategories is converted into kbinary indicator variables.

```
Encoding Summary:
  Original feature shape: (8124, 21)
  Train raw shape: (6499, 21), Test raw shape: (1625, 21)
  Train encoded shape: (6499, 115), Test encoded shape: (1625, 115)
  Target encoding: e -> 0, p -> 1
  Train class balance (p=1): 0.482
  Test class balance (p=1): 0.482
```

	cap-shape_b	cap-shape_c	cap-shape_f	cap-shape_k	cap-shape_s	cap-shape_x	cap-surface_f	cap-surface_g	cap-surface_s	cap-surface_y	cap-color_b	cap-color_c
5249	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0
5781	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	1.0	0.0	0.0	0.0
7586	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0



7.0 Conclusion

In conclusion, Phase 1 successfully transformed the mushroom dataset into a clean, analysis-ready foundation for modeling. We handled missing values and duplicates, identified rare categorical outliers using a suitable frequency-based method, and assessed feature relationships with Cramer's V and Theil's U. These steps improved data quality and clarified which variables carry the strongest signal for distinguishing edible from poisonous mushrooms.

The dataset is now properly prepared for Phase 2 through leakage-safe encoding and train-test separation, ensuring reliable model evaluation. Overall, this phase delivered a robust and reproducible preprocessing pipeline, along with clear feature insights that support confident classifier development in the next